



Unofficial FTC Discord Presents

Chain Reaction

PRESENTED BY  **gOBILDA**
DREAM • MAKE • REPEAT

Official

Competition Manual

Thank you to our sponsors!



1.1 Welcome to Chain Reaction!

This manual serves primarily as a pre-competition guide, based on the **Into the DeepSM** Competition Manual. Unless otherwise specified later in this manual or in a Team Update, all rules regarding hardware legality, most gameplay rules (i.e. blocking, pinning, etc.), pre-match procedures, and actuator limits should be derived from the **Into the DeepSM** manual. This includes, but is not limited to, legal actuators, the control system, batteries, and driver systems.

Robot construction rules should also follow the **Into the DeepSM** manual aside from section 12.1.

1.2 About this competition

Chain Reaction is an unofficial CAD challenge developed by FIRST alumni and community members from the Unofficial FIRST Tech Challenge Discord server: <https://discord.gg/ftc>

This manual serves as the guide and rulebook for the 2026–2027 Unofficial FTC Discord CAD Competition. It explains how the game is played and provides the rules teams must follow when designing a robot for the challenge.

1.3 Gracious Professionalism

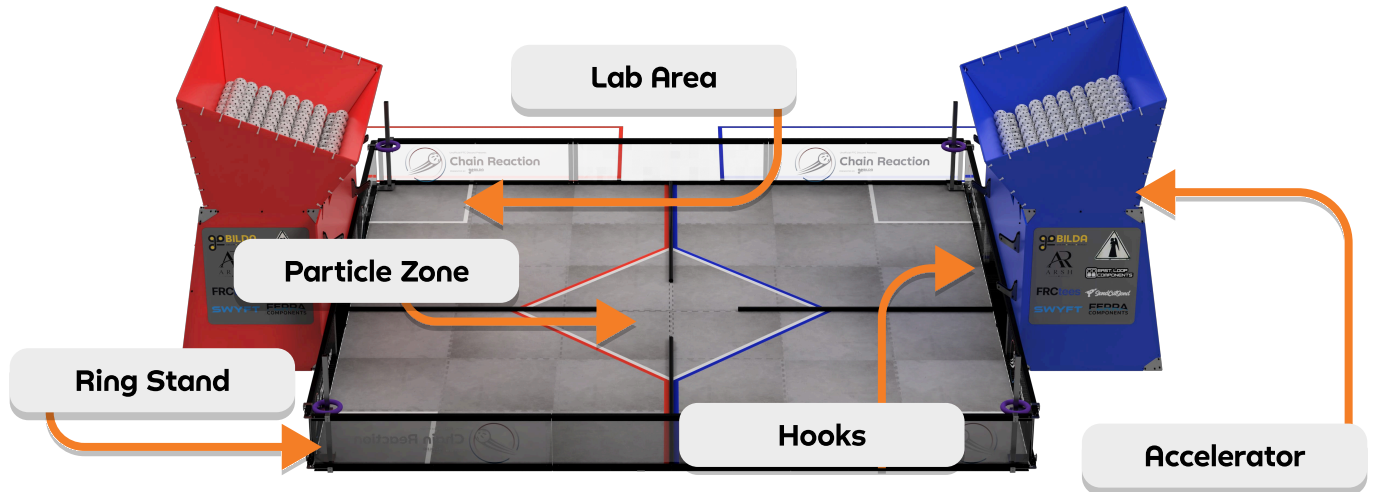
Chain Reaction is built on the belief that competition should bring out the best in people. Few ideas represent that belief more clearly than **Gracious Professionalism[®]**, a principle championed by **Dr. Woodie Flowers** and carried forward throughout the FIRST community.

Gracious Professionalism[®] means competing with passion while treating others with respect. It means striving to do excellent work without diminishing the efforts of others. It means helping another team even when they may become your opponent, sharing knowledge freely, and understanding that the way we compete matters just as much as the final result.

This competition honors Woodie Flowers by asking every participant to practice those values. Teams are encouraged to design boldly, solve difficult problems, and push themselves to improve, while also supporting the community around them. The strongest teams are not only the ones with creative designs, but the ones that make the competition better for everyone.



2. Game Introduction



2.1 Gameplay Overview

Welcome to Chain Reaction, presented by **goBILDA**!

Matches are played on a playing field, seen above. Each match features two alliances: one Red Alliance and one Blue Alliance. Each alliance is made up of two Teams.

Each Match begins with a **30-second Autonomous Period**, during which Robots operate using only pre-programmed instructions and sensor inputs. During the Autonomous Period, Robots may perform the following actions:

- Launch **Particles** into the **Accelerator** to earn points.
- Place **Catalysts** to increase the number of points earned by each **Particle**.
- Descend from the **Ring Stand** and leave the **Lab Area**.

After the Autonomous Period, the Match continues with a **two-minute Driver-Controlled Period**. During this period, Alliances may perform the following actions:

- Launch **Particles** into the **Accelerator** to earn points.
- Place **Catalysts** to increase the number of points earned by each **Particle**.
- De-score the **opposing Alliance's Catalysts** from their **Accelerator**.

The final 20 seconds of the Driver-Controlled Period are known as the End Game. During the End Game, Robots may continue performing Driver-Controlled actions and may also earn additional points by completing the following actions:

- Park in the **Lab Area**.
- Ascend the **Ring Stand**.

3. Points and Gameplay

Element	Description	Auto	Tele-Op	Endgame
 Particle	Particle scored in accelerator	1pt*	1pt*	
 Catalyst	Scored on accelerator hook	+1pt per/particle scored		
Ring Stand	Descending from the ring stand	20pts		
Ring Stand	Ascending the ring stand			20pts
Lab Area	Leaving the Lab Area	5pts		
Lab Area	Parking in the Lab Area			5pts

? Robots cannot earn both the **Ascended** and **Parked** bonus at the same time.

? There are **4 Catalysts** and **300 Particles**.

3.1 Game Automation

The Chain Reaction field features automations and auto-scoring features that interact with the game at various points.

The Accelerator features an **auto-score and reject system** that counts scored particles and launches them back onto the field. This feature is used to randomize the field pre-match after robots are initialized by randomly distributing all 300 particles across the playing area.

This automation also runs throughout the Driver Controlled period, ensuring that particles are re-distributed onto the field as they are scored.

The Hooks also feature their own **auto-score system** that confirms the Catalyst has been successfully placed and ensure that the scoring bonus to Particles is applied correctly.

? The abuse of or intentional manipulation of automated scoring features on the field is grounds for a red card at the discretion of the head referee.

3.2 Scoring

3.2.1 Ascended and Descended status are defined by Robot support.

A Robot is considered **Ascended** when it is supported only by the vertical structure of the Ring Stand.

A Robot is considered **Descended** when it is partially or fully supported by a surface other than the vertical structure of the Ring Stand.

A Robot is considered **Parked** in the Lab Area when it is completely in the the outer edge of the Lab Area tape.

A Robot has **Left** the lab area when it is completely outside of the Lab Area.

? A Robot touching the tile floor is not **Ascended**, even if part of the Robot is also touching the Ring Stand.

3.2.2 Robots may not contact the top surface of the Ring Stand.

Robots may use the vertical structure of the Ring Stand for Ascending, but may not use the top surface or mounting brackets as a platform, brace, hook point, or support surface.

3.2.3 Catalysts and Particles are scored automatically.

Particles are scored once they pass through the automated score and reject system.

Catalysts are scored once they are placed onto a hook and successfully press the switch inside the hook's structure.

3.3 Gameplay Rules

G01 Robots may control a maximum of one Catalyst at a time.

A Robot may Control no more than one Catalyst at any time. Robots may Control an unlimited number of Particles at a time, provided no other rule is violated.

? Control includes carrying, holding, trapping, herding, or intentionally moving a Scoring Element for strategic advantage. Brief or accidental contact with additional Catalysts is not considered Control.

G02 There are control area limits for Particles.

Robots may not control Particles outside of an 18 in. x 24 in. x 18 in. tall. rectangular prism.

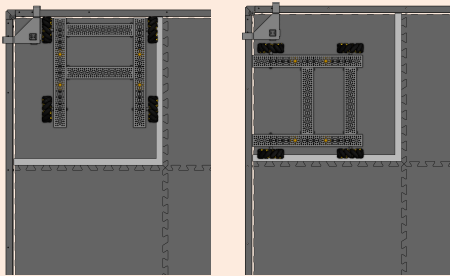
? **Momentary Control** (i.e. an intake pulling in a particle or contacting it briefly) and **Unintentional Control** (i.e. a particle landing briefly on top of a robot, plowing) are acceptable.

G03 Robots have expansion limits.

During inspection and the pre-Match period, each Robot must fit within an 18 in. x 18 in. x 18 in. starting configuration.

After the start of the Match, Robots may expand, but must remain within a 24 in. wide x 24 in. long x 30.5 in. tall rectangular prism.

G04 Robots must begin the match completely in the lab area.



✓ Legal Lab Area Examples

A Robot may start either on the floor of the **Lab Area** or **Ascended on the Ring Stand**. In both cases, the Robot must still be completely in the **Lab Area** at the start of the Match. Teams should make their starting configuration easy for referees to verify before the Match begins.

At the start of the Match, each Robot must be fully contained within its Alliance's Lab Area.

Robots may begin the Match in one of the following starting configurations:

- Fully supported by the tile floor and completely within the Lab Area and touching one of the field walls.
- Ascended on the Ring Stand

? The post-start expansion limit applies at all times after the Match begins, including Autonomous, Driver-Controlled, and End Game periods.

G05 Robots may not contact an opposing robot that is Ascending.

During the End Game period, Robots may not contact an opposing Robot that is in contact with a Ring Stand. Doing so results in a major foul per offense.

G06 Contact during Autonomous has specific restrictions.

During the Autonomous Period, the Red Alliance Section and Blue Alliance Section are protected. A Robot that contacts an opposing Robot that is completely within its Alliance Area receives a Major Foul.

The Particle Zone is not protected. Robots in the Particle Zone forfeit all Alliance Area protections, and contact involving Robots in the Particle Zone does not result in a Foul solely because of that contact.

G07 Scored Catalysts are not protected.

Alliances may de-score Catalysts scored by the opposing Alliance.

- De-scoring an opposing Catalyst is legal gameplay. However, Robots must still follow all other rules, including rules about Field damage, Robot contact, expansion, and unsafe behavior.

G08 Robots may not restrict Field or Scoring Element access for prolonged periods.

Robots may not use Field position, Scoring Elements, or Robot mechanisms to restrict access to important Field areas or Scoring Elements for a prolonged period.

- Defense is allowed, but strategies that prevent normal gameplay access for an extended period are not. Teams should avoid strategies that make Scoring Elements or Field Elements effectively unavailable to the opposing Alliance.

G09 Robots must not obstruct the Accelerator.

Robots may not obstruct the Accelerator exit area. Obstructing the Accelerator exit area results in one Major Penalty for each Particle blocked.

G10 The game ends at the end of the buzzer.

Buzzer beaters return! Endgame scoring is done at the end of the buzzer.

3.3 Additional Vocabulary

3.3.1 Red/Blue Alliance Sections

The Red and Blue Alliance sections are defined as columns A-C and D-F respectively (columns viewed from audience side), excluding the particle zone.

3.3.2 Particle Zone

The Particle zone is the area in the diamond of white tape in the field's center, including the tape.

4. Field and Element Specifications

Catalyst Specifications

Dimensions:

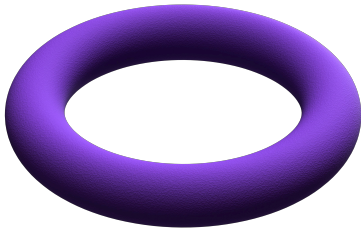
- 1 in. Thickness
- 6 in. OD

Material:

- Injection Molded ABS

Weight:

- ~45g



Particle Specifications

Dimensions:

- 3 in. OD

Material:

- Polyethylene Plastic

Weight

- ~21g



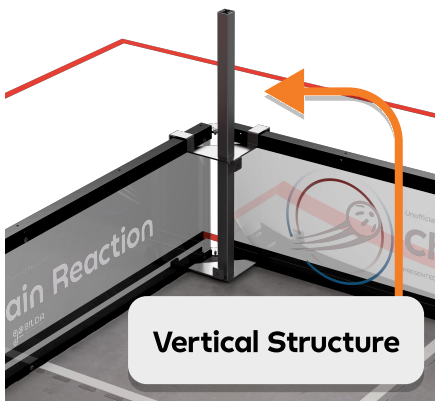
Ring Stand Specifications

Dimensions:

- 22.5 in. tall Vertical Structure

Material:

- Steel



Accelerator Specifications

Dimensions:

- 47 in. tall front lip
- 30.5 in. tall top hook
- 19.5 in. bottom hook

Material:

- Hooks: aluminum (6061)
- Panels: corrugated plastic
- Accelerator ramp: polycarbonate

